HIMS-AI-2526

AI Powered Hospital Information Management System

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Department of Computer Science — Capstone Project

Presentation Agenda

Project Overview

⚠ Problem

Hospitals face manual data entry, fragmented records, and administrative overload — leading to delays and errors in patient care.

AI Recording

Real-time doctor-patient conversation capture & transcription

NLP Extraction

GPT-based extraction of symptoms, diagnosis & medications

Doctor Verification

Doctors review AI notes before they are saved to the database

Smart Dashboards

Role-based dashboards for Doctors, Admin, Receptionists & Patients

Project Scope — Inclusions & Exclusions

✓ AI-assisted medical note creation from doctor-patient conversations
✓ Smart appointment & real-time queue management
✓ Controlled patient self-registration (staff-verified activation)
✓ Centralized patient record & prescription management
✓ Integrated staff dashboards for all user roles
✓ Secure messaging, billing, and downloadable reports
✓ Healthcare analytics and operational insights

✗ No autonomous disease diagnosis without doctor approval
✗ No cross-hospital automatic data transfer
✗ No payroll or pharmacy stock management
✗ No emergency dispatch or ambulance coordination
✗ No predictive disease analytics in current version
✗ Language limited to English, Urdu, and Punjabi

User Roles & Characteristics

Doctor

Primary user. Manages consultations, reviews AI-generated SOAP notes, confirms and stores medical records. Requires minimal interface interruption.

Receptionist

Assigned to one doctor. Handles patient registration, appointment scheduling, queue management, and check-ins.

Admin

Manages user accounts, monitors data flow, generates analytical reports, and ensures system compliance and security.

Patient

Tertiary user. Accesses visit history, prescriptions, and reports through patient portal. Staff can assist non-technical patients.

Doctor Consultation Session Flow

1

Start Session

Opens fullscreen session modal; loads patient history & SOAP template; requests mic permission

2

Record Conversation

Audio captured in real time; 'Recording...' pulse indicator; stop to save audio blob

3

AI Processing

Deepgram converts audio to text; Open AI GPT 4.1 extracts symptoms, diagnosis, medications

4

Review SOAP Notes

AI-generated summary displayed; doctor can edit fields; unsaved changes are highlighted

5

Submit & Store

SOAP notes + audio uploaded; appointment marked 'Completed'; records saved to PostgreSQL

Operating Environment & System Constraints

Web-Based Platform

Accessible via Chrome, Edge, Firefox on desktop or mobile devices.

Cloud Hosting

Secure cloud infrastructure for high availability, backups, and scalability.

Hardware Needs

Min. 8 GB RAM, stable internet. Django backend + ReactJS frontend.

Database & APIs

PostgreSQL for data. REST APIs for AI, NLP, and auth integration.

Software Constraint

Depends on Whisper AI & spaCy APIs; service interruptions affect accuracy.

Language Constraint

Supports English & Urdu only. Accents/dialects may reduce accuracy.

Privacy & Legal

HIPAA-equivalent compliance required. Consent needed for recordings.

Environmental

Background noise in busy hospitals can reduce transcription accuracy.

Functional Requirements

FR-01

Patient Registration

Staff-verified registration with unique hospital ID assignment

FR-02

Appointment Scheduling

First-come-first-serve booking with real-time queue tracking

FR-03

Electronic Medical Records

Secure, centralized storage of diagnoses & treatment histories

FR-04

Billing & Payment

Auto-generated bills; JazzCash / Easypaisa online payment support

FR-05

Role-Based Auth

Secure login with RBAC for Admin, Doctor, Receptionist, Patient

FR-06

Analytics & Reports

Graphical dashboards; PDF export of hospital analytics

FR-07

Notifications & Alerts

Appointment reminders, billing alerts via dashboard notifications

FR-08

Backup & Recovery

Scheduled automated backups; admin-triggered data restoration

Non-Functional Requirements

Performance

- Patient records retrieved in < 20 sec
- Appointments scheduled in < 5 sec
- Supports 100 concurrent users
- UI loads in < 6 sec on hospital network

Security

- Role-Based Access Control (RBAC)
- SSL/TLS encryption at rest & in transit
- Audit logs of all user activities
- Session timeout for inactive users

Safety

- Prevent unauthorized modification of records
- Daily automatic data backups
- Auto-recovery on server crash / power outage
- Warning messages for invalid operations

Documentation

- User manual for all roles
- Admin guide for config & maintenance
- Troubleshooting guide
- In-system help tooltips & FAQs

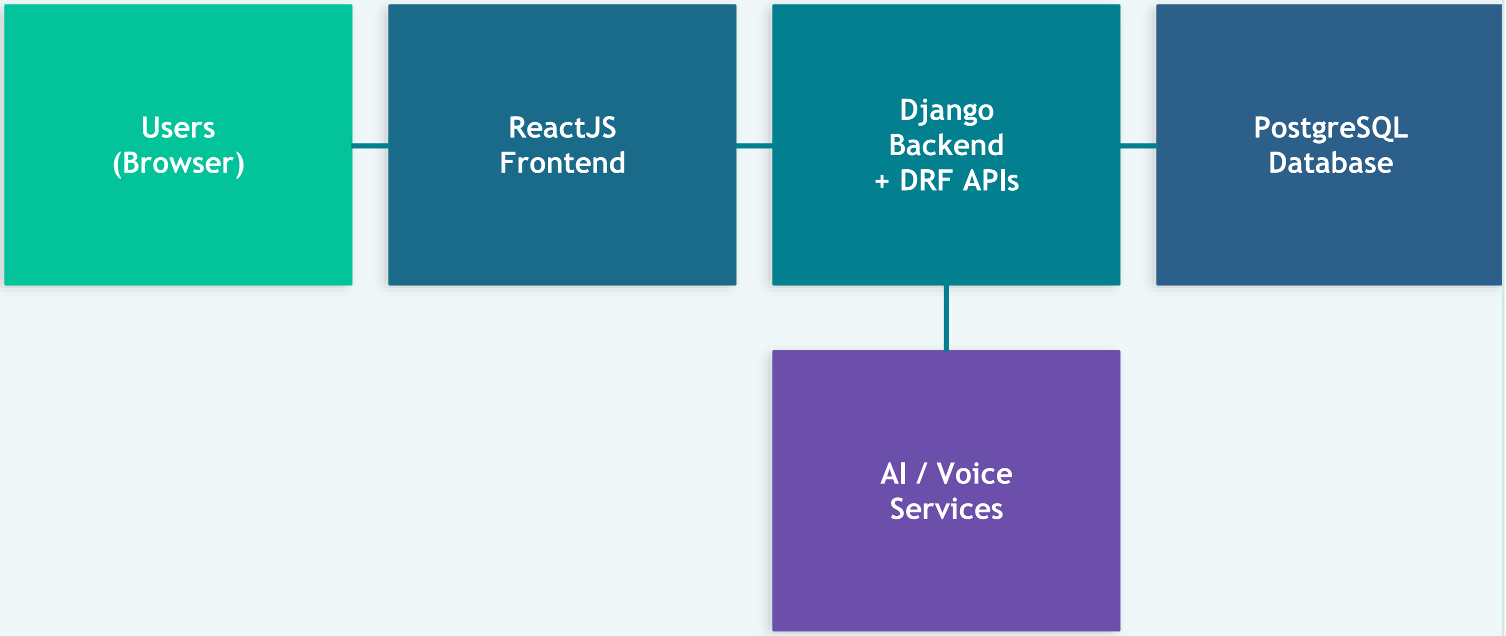
Use Cases Overview

UC-21	**AI-Assisted Medical Documentation** *Actor(s): Doctors*	Doctor starts session → AI captures conversation → NLP extracts key info → Doctor reviews & confirms → Records saved to database
UC-22	**Appointment & Queue Management** *Actor(s): Patients, Receptionists*	Patient requests appointment → Receptionist confirms → Queue position assigned → Real-time queue updates displayed
UC-23	**Patient Record Access** *Actor(s): Doctors, Patients, Receptionists*	User logs in → Searches patient → Views history, prescriptions, reports → Can download PDF reports
UC-24	**Prescription Management** *Actor(s): Doctors, Patients*	Doctor creates/updates prescription → Stored in database → Patient accesses digital prescription via portal
UC-25	**Secure Communication** *Actor(s): All User Roles*	Users send healthcare messages → System delivers securely → Dashboard alerts for appointments & billing

Technology Stack

Frontend	ReactJS | HTML5 | CSS3 | Tailwind CSS
Backend	Django (Python) | Django REST Framework | Cookie-Based Auth
Database & Storage	PostgreSQL | Cloud / Local Storage | CSV Export
AI & Voice	Deepgram| OPEN AI GPT 4.1|
DevOps & Tools	Git / GitHub | Netlify (Frontend) | Linux Server (Backend)

System Architecture Overview



- ReactJS frontend provides role-based dashboards communicating via REST APIs
- Django backend handles business logic, RBAC, AI documentation, and DB operations
- PostgreSQL stores all patient records, appointments, prescriptions, and logs
- AI / Voice modules process speech-to-text and NLP for structured medical notes
- External SMTP / email APIs support notifications and appointment alerts

Key Module Implementations

AI Consultation & Documentation

Voice capture → Deepgram→ Open AI GPT → SOAP note generation → Doctor confirmation → PostgreSQL storage

Communication & Notification

Secure healthcare messaging; appointment reminders and billing alerts via dashboard and email

Appointment & Queue Management

Patients book online → System assigns FCFS queue position → Real-time status updates via dashboard

Administrative Dashboard

User management, doctor-receptionist assignments, analytics dashboards, billing, and activity monitoring

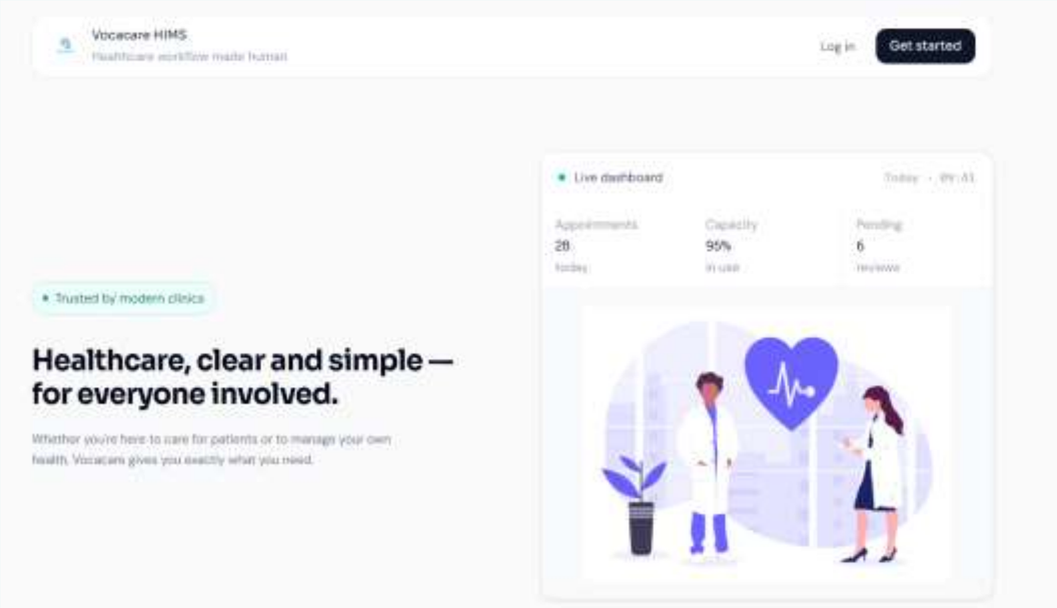
Patient Record & Prescription

Centralized records with prescription history; downloadable PDF reports for patients and doctors

Authentication & RBAC

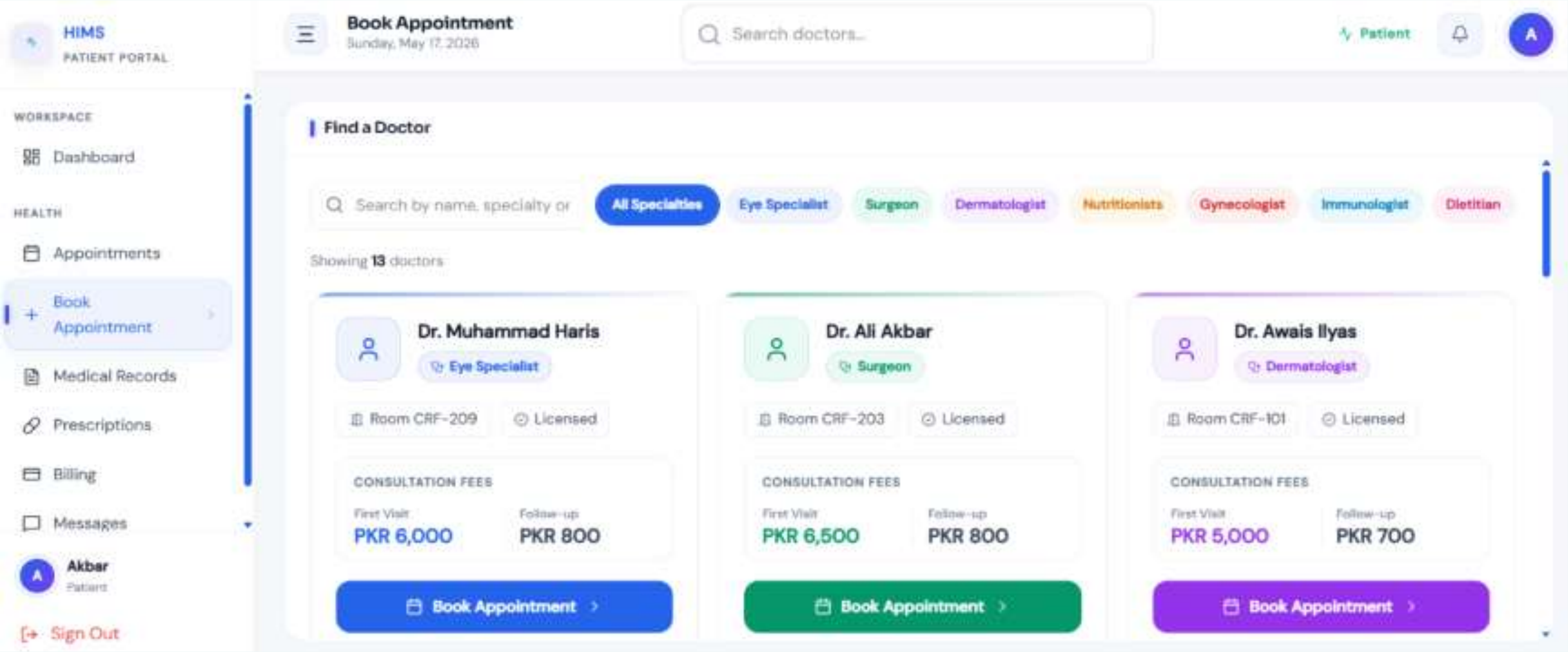
Role-based access control for Admin, Doctor, Receptionist, Patient; cookie-based session management

UI Interfaces — Dashboards



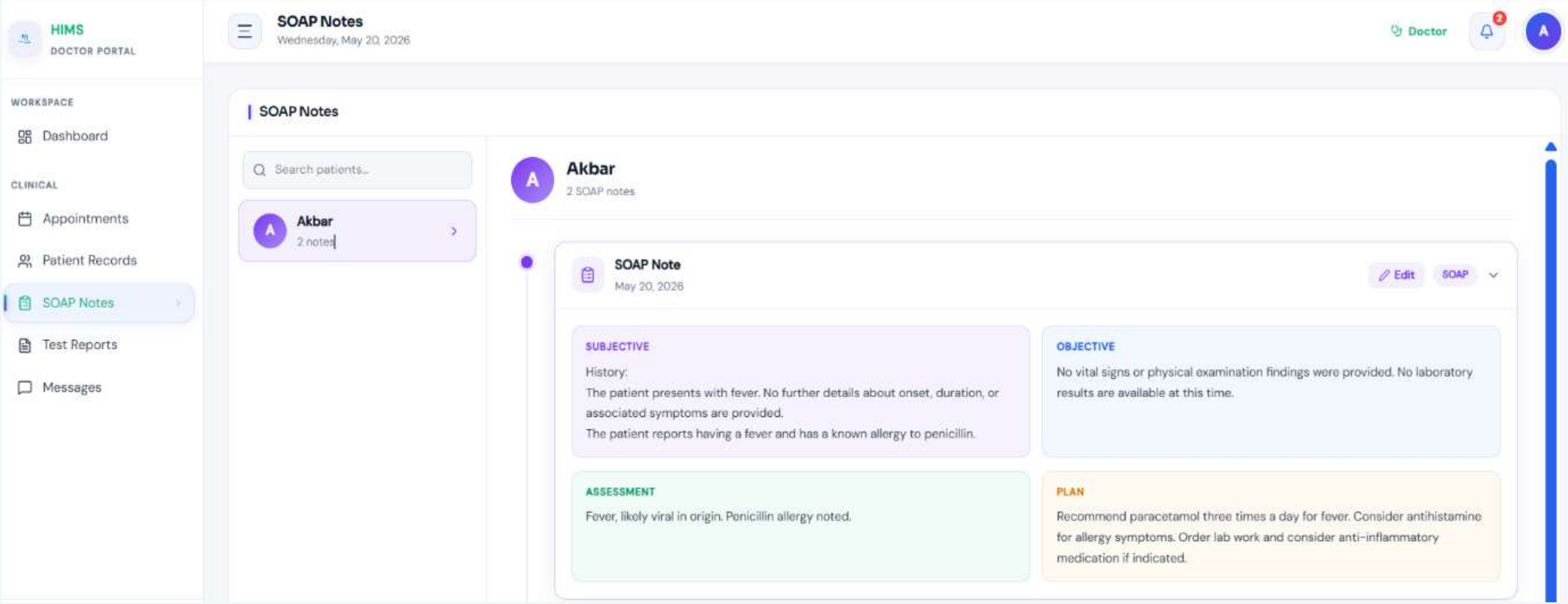
Landing Page

UI Interfaces — Dashboards



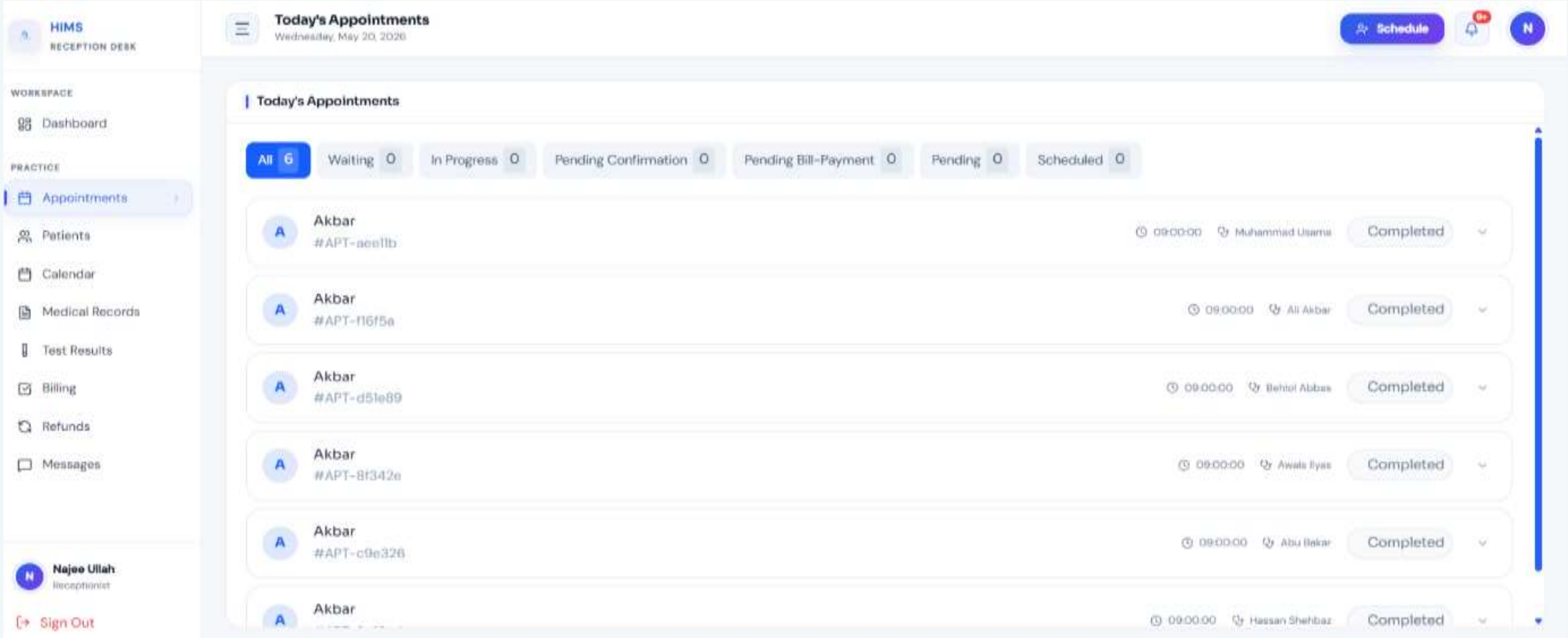
Patient Dashboard

UI Interfaces — Dashboards



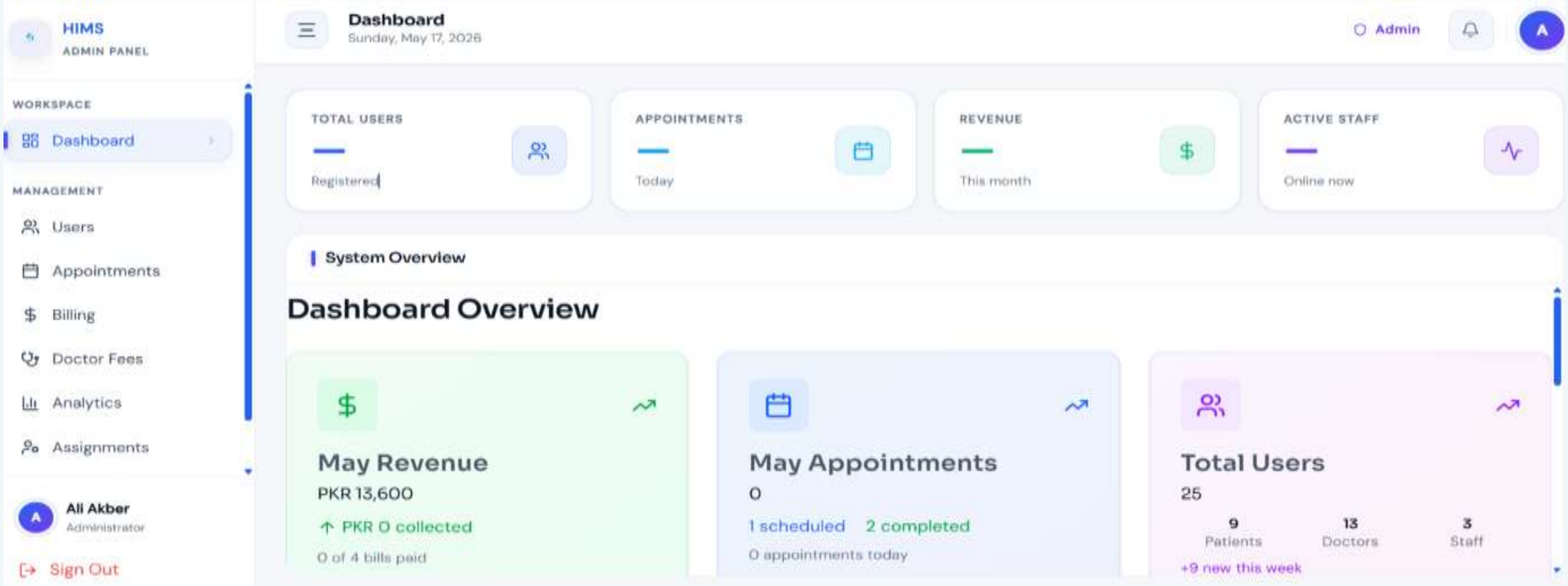
Doctor Dashboard

UI Interfaces — Dashboards



Receptionist Dashboard

UI Interfaces — Dashboards



Admin Dashboard

Functionality & Performance Testing

Functional Testing

- Patient registration, login, role access verified
- Appointment booking and real-time queue updates
- AI recording → transcription → SOAP note → storage
- Prescription creation, access, and PDF download

Performance Testing

- Record retrieval < 20 sec under load testing
- 100 concurrent users simulated — no degradation
- UI load time < 6 sec on standard network
- Report generation for 5,000 records < 7 sec

Security Testing

- RBAC: unauthorized access blocked and logged
- Encryption verified for data at rest and in transit
- Session timeout enforced for inactive users
- SQL injection and XSS protection tested

Conclusion & Impact

AI-Driven Documentation

Eliminates manual data entry; SOAP notes generated automatically from doctor-patient conversations.

Operational Efficiency

Real-time queue management and centralized records reduce waiting times and administrative load.

Secure & Compliant

RBAC, SSL/TLS encryption, audit logs, and HIPAA-equivalent compliance protect patient data.

Scalable Architecture

Modular design supports future AI enhancements and integration with additional hospital systems.

"Transforming conversations into care."

Thank You